

Member 1 Handoff: Data Preparation and EDA

NYC Street Tree Health Predictor | Intro to Data Science Final Project | June 27, 2026

Role Summary

Member 1 is responsible for the project framing, cleaned dataset, feature engineering for EDA/modeling, data visualization, and data limitation notes. This package gives the team ready-to-use files for the final six-page Streamlit app.

Dataset Snapshot

Item	Value
Dataset	NYC Open Data: 2015 Street Tree Census - Tree Data
Target	health: Good, Fair, Poor
Cleaned rows	49,994
Main predictors	species, borough, diameter, stewardship, guards, sidewalk, visible problems

Cleaning and Feature Engineering

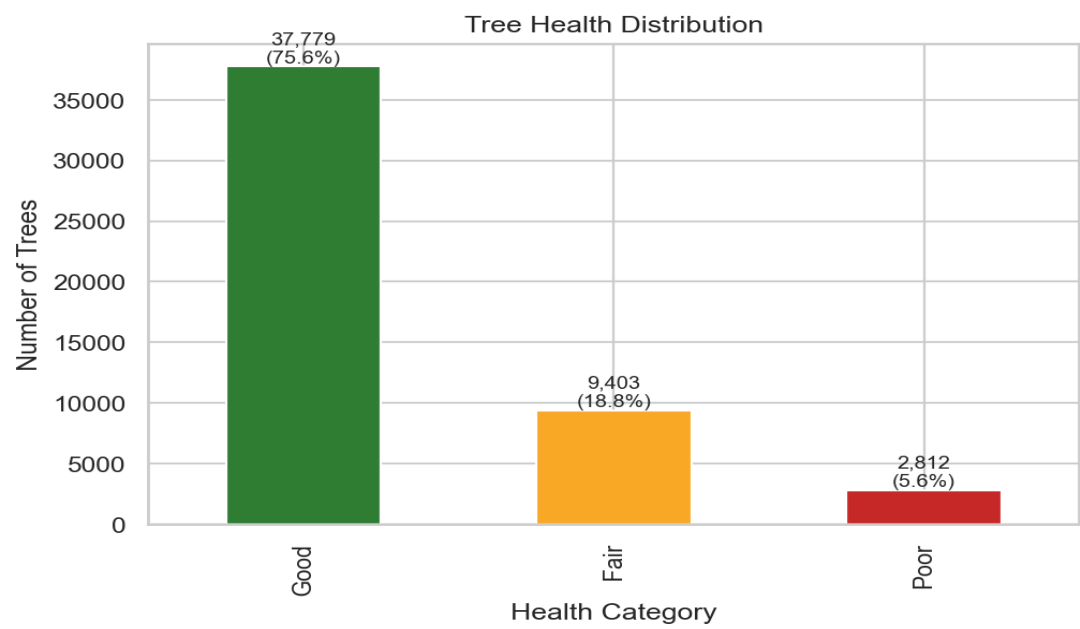
- Dropped rows missing health, tree diameter, borough, or species.
- Standardized missing categorical values to Unknown.
- Filtered unrealistic tree_dbh values outside 0-100 inches.
- Created problem_count, has_problem, dbh_group, species_top15_or_other, and binary problem flags.

Key Findings

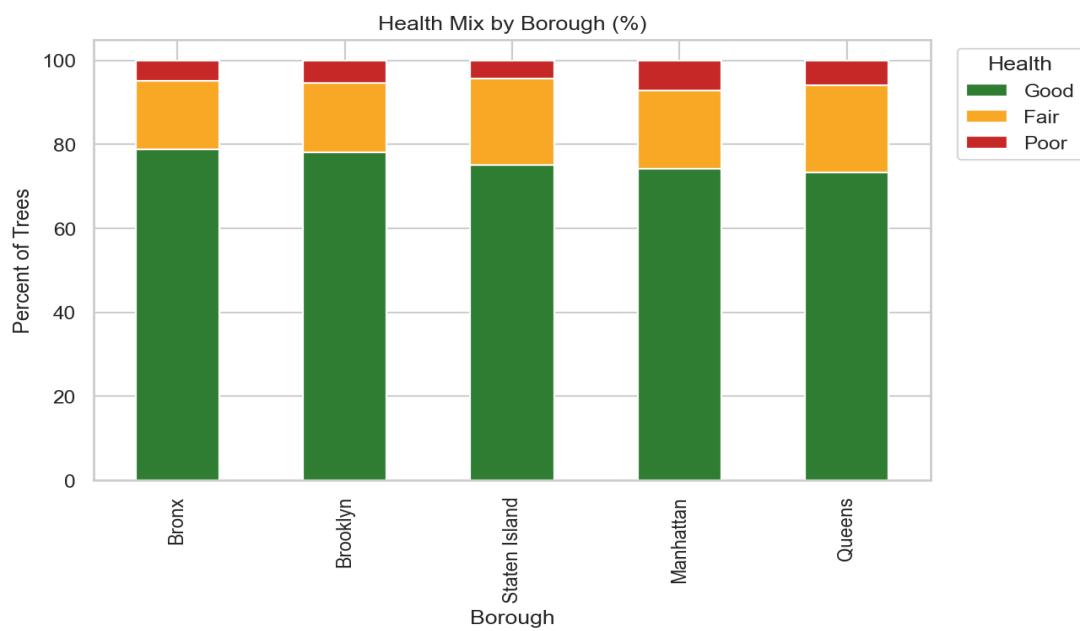
- The cleaned sample contains 49,994 usable tree records after dropping missing target/core fields.
- Health distribution: Good 75.57%, Fair 18.81%, Poor 5.62%.
- The largest species groups are: London planetree, honeylocust, Callery pear, pin oak, Norway maple.
- Engineered features added for app/modeling: problem_count, has_problem, dbh_group, species_top15_or_other, and binary root/trunk/branch problem flags.
- Recommended modeling note: because the target is imbalanced, compare models with macro-F1 plus a confusion matrix, not accuracy alone.

EDA Visuals

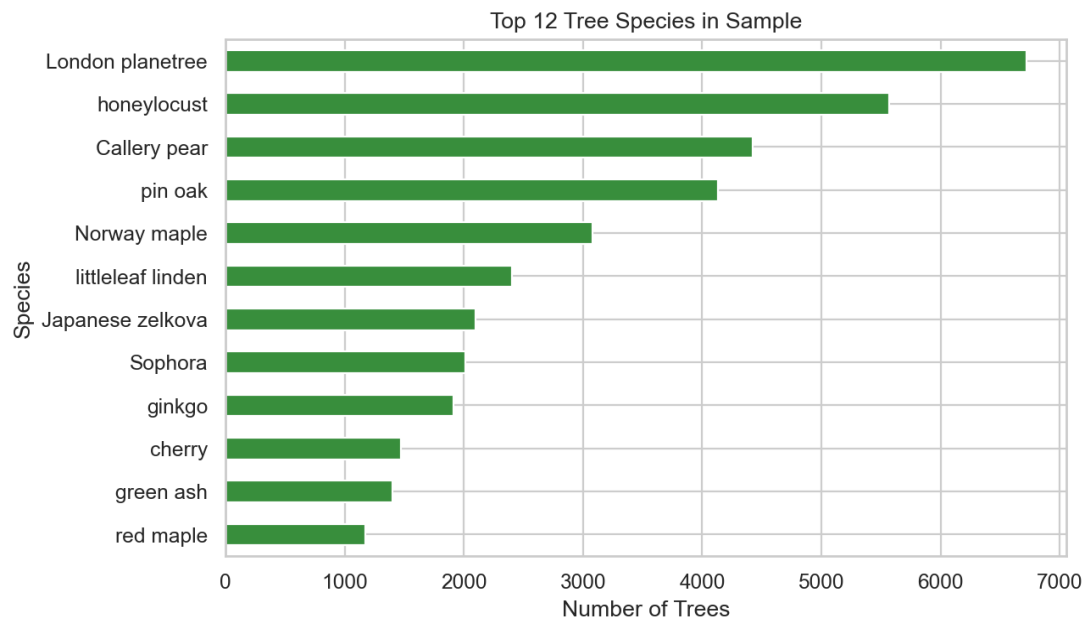
01 Health Distribution



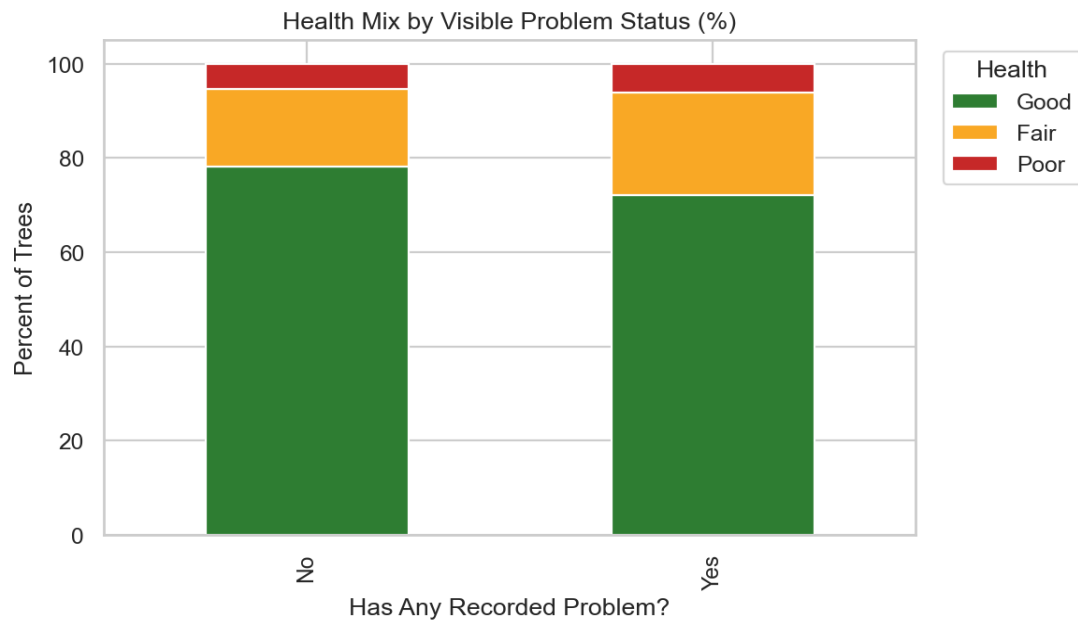
02 Health By Borough Pct



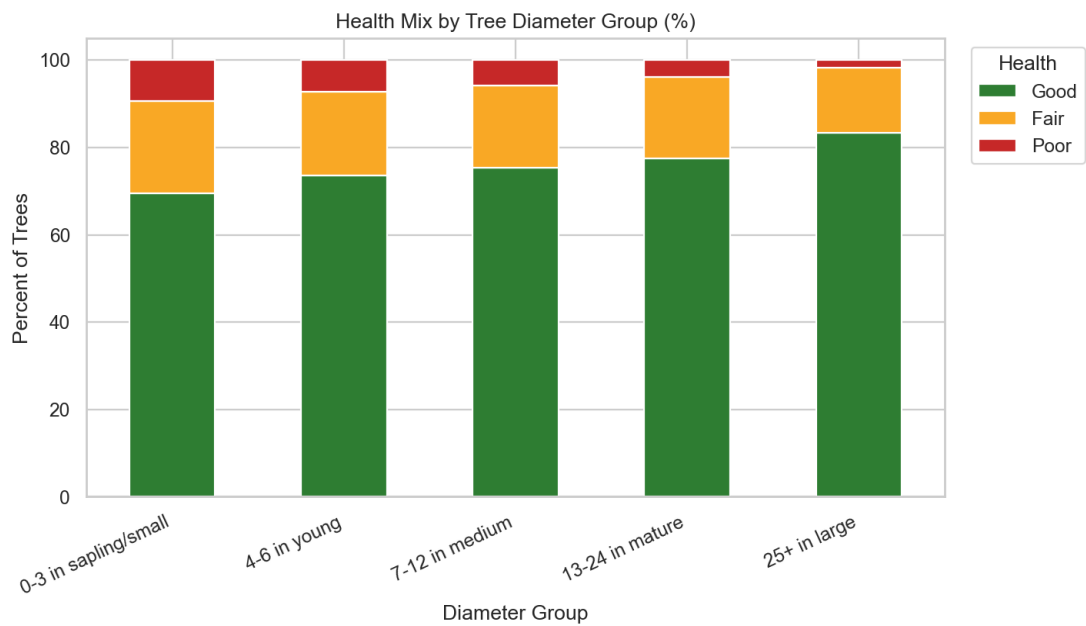
03 Top Species



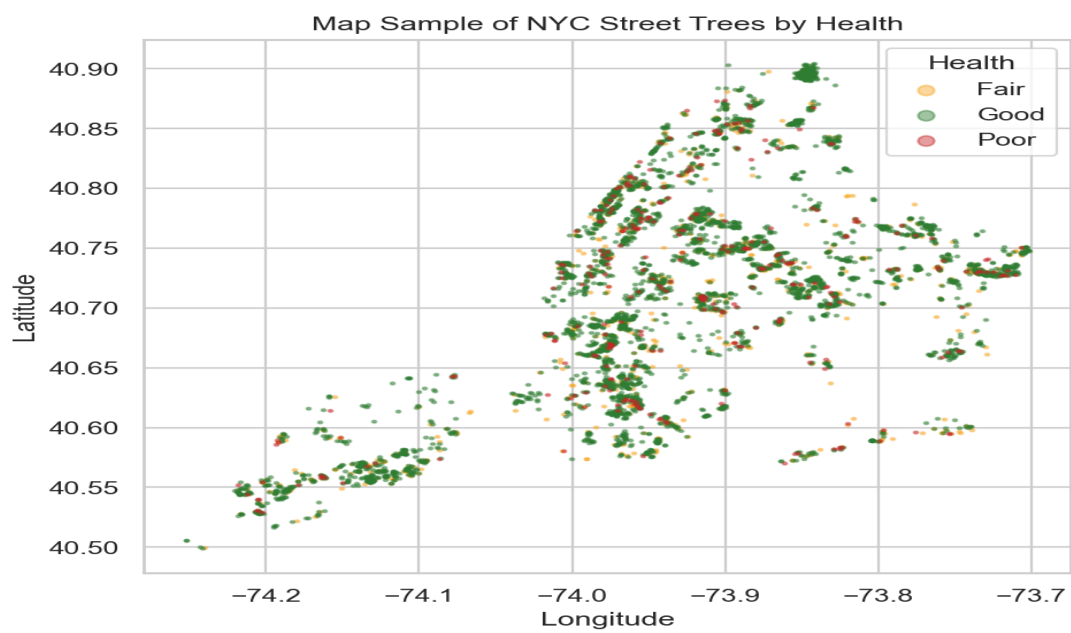
05 Health By Problem Status



06 Health By Dbh Group



07 Map Sample By Health



Team Handoff

Member 2 should use `data/nyc_tree_member1_model_ready_sample.csv` for model comparison. Member 3 should copy the data folder and Streamlit page files into the final app repo, then deploy the completed app on Hugging Face Spaces.

Required final app pages: Introduction, Data Visualization, Multiple Selectable Models, Feature Importance/Explainability, Hyperparameter Tuning/Experiment Tracking, and Conclusion.

Limitations

- Target classes are imbalanced, so macro-F1 and confusion matrices are important.
- The data is from 2015, not a live monitoring feed.
- The model can support triage and learning but should not be treated as an official maintenance decision system.
- Observed problem fields may include reporting or inspection bias.